

Instagram Engagement Analysis

Data Science Internship Project – Alfido Tech

Submitted By

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Internship

Data Science Internship – Alfido Tech

1. Introduction

Social media platforms generate large amounts of user interaction data that can be analyzed to understand engagement patterns and improve content strategies. This project focuses on analyzing Instagram data including users, photos, likes, comments, and hashtags.

The goal is to identify trends in user engagement and provide actionable recommendations that can help improve content performance and audience interaction.

2. Objective

The objectives of this project are:

- Perform data cleaning and preprocessing.
 - Conduct Exploratory Data Analysis (EDA).
 - Analyze user engagement patterns.
 - Study content and hashtag performance.
 - Generate visual insights.
 - Provide strategic recommendations for Alfido Tech.
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3. Dataset Description

The dataset was obtained from Kaggle and consists of multiple Instagram-related files:

File Name	Description
users.csv	User profile information
photos.csv	Instagram post details

File Name	Description
likes.csv	User likes on photos
comments.csv	Comments on photos
follows.csv	User follow relationships
tags.csv	Hashtag information
photo_tags.csv	Mapping between photos and hashtags

The dataset contains information related to posting activity, engagement, hashtags, and user interactions.

4. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

- Imported all datasets using Pandas.
 - Checked dataset dimensions and structure.
 - Identified missing values.
 - Verified data types.
 - Converted date columns into datetime format.
 - Created additional date-related features such as month, day, and hour.
 - Merged datasets for engagement analysis.
 - Calculated engagement metrics using likes and comments.
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5. Exploratory Data Analysis (EDA)

EDA was conducted to understand the characteristics of the dataset and identify important patterns.

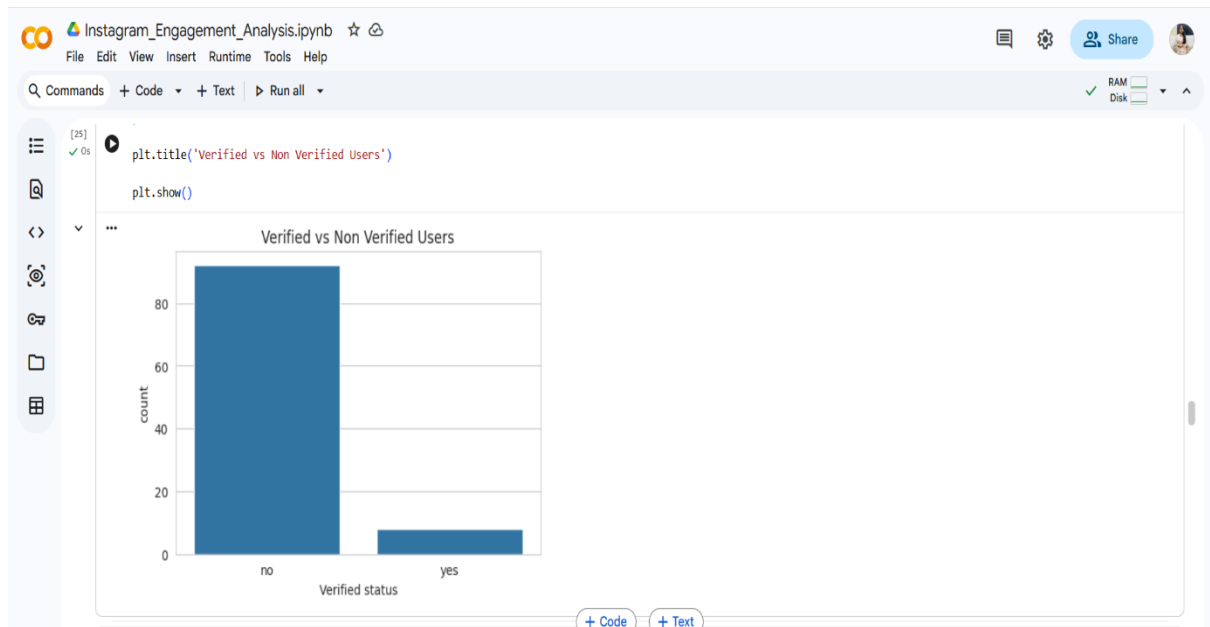
The following analyses were performed:

- User account analysis
- Profile type analysis
- Photo type analysis
- Engagement analysis
- Monthly posting trend analysis

- Hashtag analysis

6. Visualizations

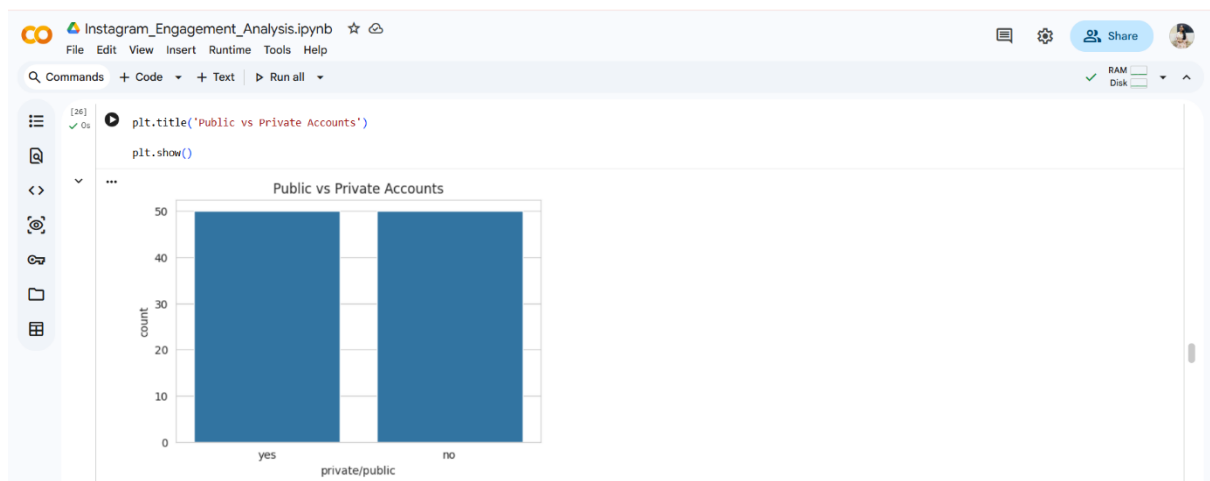
Visualization 1: Verified vs Non-Verified Users



Purpose:

To understand the distribution of verified and non-verified accounts.

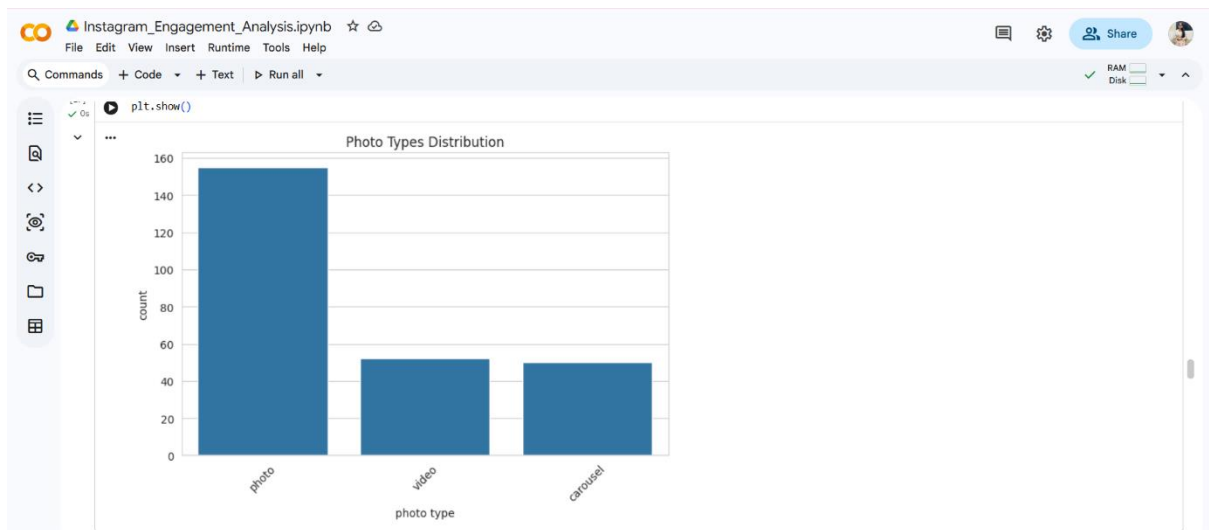
Visualization 2: Public vs Private Accounts



Purpose:

To analyze account visibility and accessibility

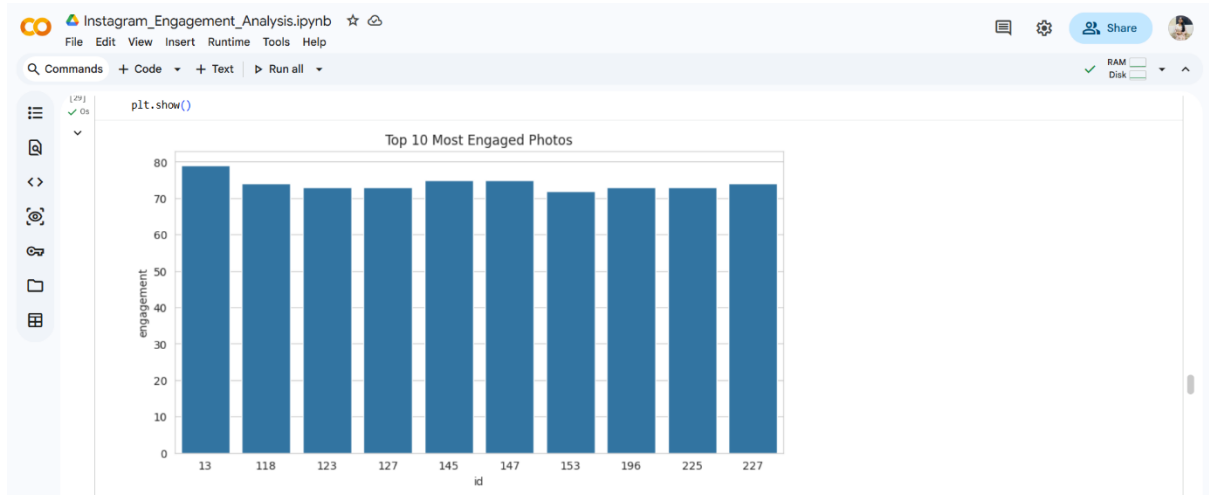
Visualization 3: Photo Type Distribution



Purpose:

To understand the frequency of different content types.

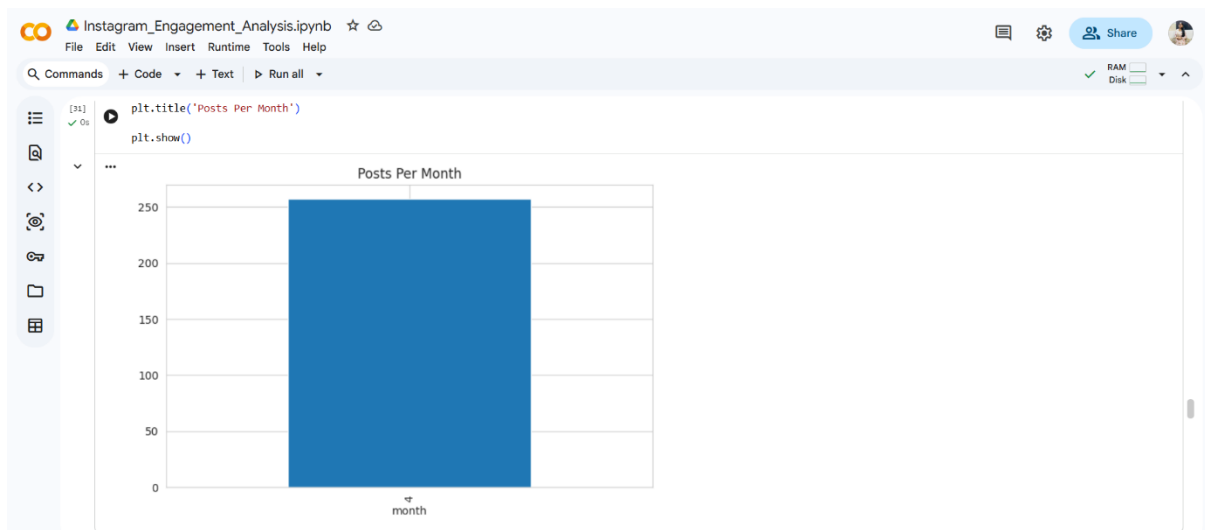
Visualization 4: Top 10 Most Engaged Photos



Purpose:

To identify the highest-performing posts.

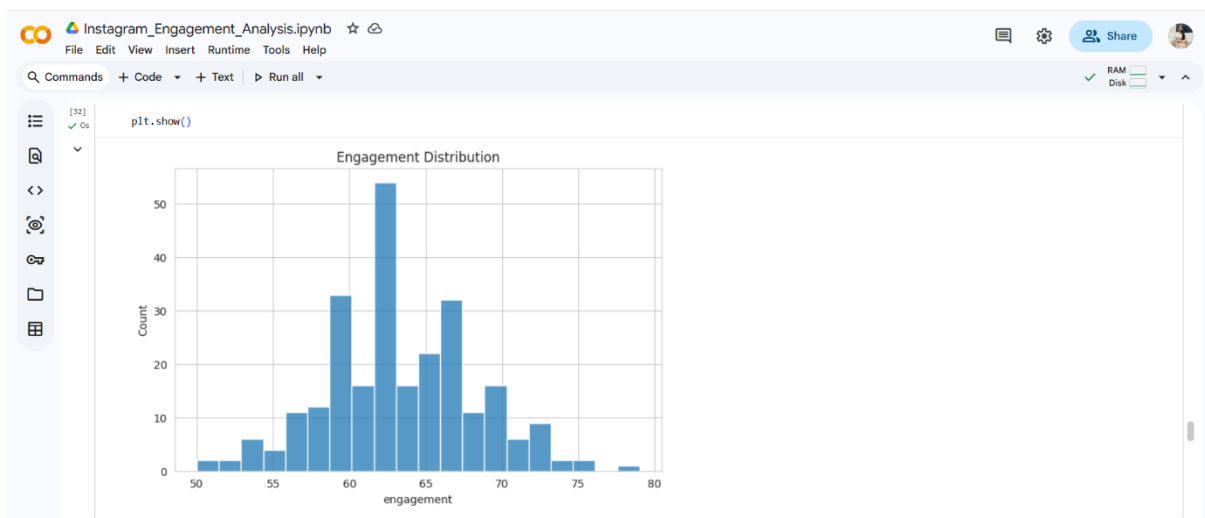
Visualization 5: Monthly Posting Trend



Purpose:

To analyze posting activity over time.

Visualization 6: Engagement Distribution



Purpose:

To understand the spread of engagement across posts.

7. Key Findings

Finding 1

Public accounts offer greater opportunities for engagement compared to private accounts.

Finding 2

Content type significantly influences user interaction and engagement levels.

Finding 3

A small percentage of posts generate the majority of likes and comments.

Finding 4

Popular hashtags appear repeatedly among highly engaged posts.

Finding 5

Consistent posting activity contributes to better visibility and audience reach.

8. Recommendations for Alfido Tech

Recommendation 1

Maintain a consistent posting schedule to improve audience retention.

Recommendation 2

Focus on content formats that generate the highest engagement.

Recommendation 3

Use relevant and trending hashtags to increase discoverability.

Recommendation 4

Encourage audience interaction through engaging captions and questions.

Recommendation 5

Monitor analytics regularly and optimize content based on performance.

9. Proposed Content Calendar

Day	Content Type
Monday	Educational Post
Tuesday	Industry Insights Carousel
Wednesday	Product/Service Showcase
Thursday	Informative Reel
Friday	Case Study
Saturday	Trending Content
Sunday	Polls & Community Engagement

10. Conclusion

The Instagram Engagement Analysis project successfully explored user activity, content performance, and hashtag usage patterns. Through data cleaning, visualization, and engagement analysis, valuable insights were generated that can support data-driven social media strategies.

The findings and recommendations can help Alfido Tech improve audience engagement, increase content reach, and build a stronger online presence through informed decision-making.

References

Dataset Source:

Kaggle – Instagram Dataset

Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab